Power-board battery case comparison

Assembly fit, bolt tightening, side pull and vibration response

9 November 2026 | Native structural analysis in ANSYS MAPDL 2026 R1.02

The mounting height determines whether the cells are already loaded

Placing the four tabs directly against the board creates approximately 1.18 mm of interference at the cell crowns. Each of the three cell solids shares about 1.70 cm^3 with the undeformed enclosure. A 3 mm tab stand-off clears the cells, nickel strips, washers and fasteners present in the 35-body mechanical STEP assembly. Thus a flush installation must displace or deform parts as the screws tighten. During rigid installation from the raised position, the four outer washers touch first while 2.397 mm of tab stand-off remains. Translating a rigid cell along the PCB X axis first closes a gap after 0.146–0.325 mm, depending on the cell and direction. Flexible relative motion determines whether contact occurs under load.

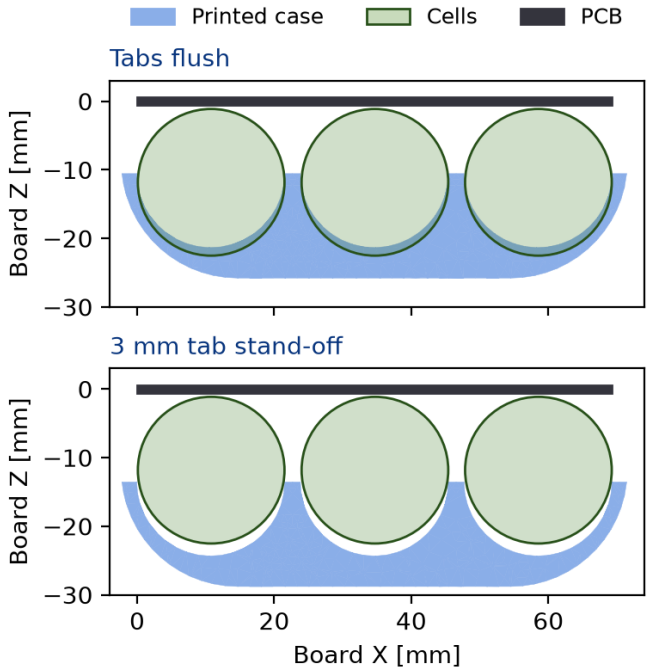


Figure 1: Section through the middle of the cells, in PCB coordinates. The solid outlines show as-designed geometry before tightening. The mounting tabs lie beyond this section.

What the raised case changes

Under X-axis vibration, the 3 mm case increased peak board-normal motion from 23.37 μm RMS bare to 145.55 with nylon screws and 120.00 with aluminium. Under base Z the peak PCB stress rose from 32.5 MPa RMS bare to 64.8 with nylon and 84.2 with aluminium (page 13). Nylon gave the lowest cell-body X acceleration. Aluminium reduced board movement relative to nylon and increased the local stress across the PLA layers at a screw bearing. In the whole-board flush trial with nylon screws, all four tabs began touching the PCB at 2.60–2.675 mm of screw travel. At 2.75 mm, about 95% of each tab footprint still had more than 0.05 mm clearance (page 19).

Reference assembly

The October assembly has 208 bodies and a modeled mass of 251.82 g. Eight stand-off end faces carry the base motion. Two nickel strips attach each cell to the assembly. In the connected reference, six existing screws were loaded to 750 N apiece and locked. Its first mode was 485.964 Hz. Independent base excitation between 20 and 2000 Hz gave a maximum PCB directional displacement of 0.052674 mm RMS under Z excitation and a maximum PCB equivalent stress of 90.325 MPa RMS under X excitation. These archived quantities supply reference checks for the repeats.

Registration and fastener selection

The four nonplated corner holes in PCB1 ODB++ have finished diameter 4.200 mm. The hole pattern measures 62.865×114.427 mm. The case bores have two diameters, 5.000 and 5.090 mm, on a 62.855×113.241 mm pattern. Centring the two patterns leaves about 0.593 mm of offset at each end. The sum of the radial clearances of a nominal 4 mm shank in the smaller case bore and in the board is 0.600 mm. The approximately 0.007 mm difference is the available nominal alignment margin there. Printed hole size, fastener major diameter and drilling tolerances therefore matter for installation.

Tab stand-off [mm]	Cell overlap [mm^3/cell]	Washer bodies intersected	Other intersections
0	1696–1698	4	6 nickel strips
1.2715	0	4	None found
2	0	4	None found
3	0	0	None found

Table 1: Solid intersections between the unstrained case and mechanical STEP bodies. The board's lower face defines the tab seating datum.

Starting from 3 mm of tab stand-off, the four outer washers meet the descending case after 0.603 mm. The middle cell first touches after 1.760 mm and the two outer cells after 1.790 mm. Once a washer touches, actual motion depends on deformation of the case and fasteners. A rigid assembly at the nominal registration therefore needs more than 2.397 mm of stand-off for clearance. Manufacturing tolerances and assembly misalignment would change that value.

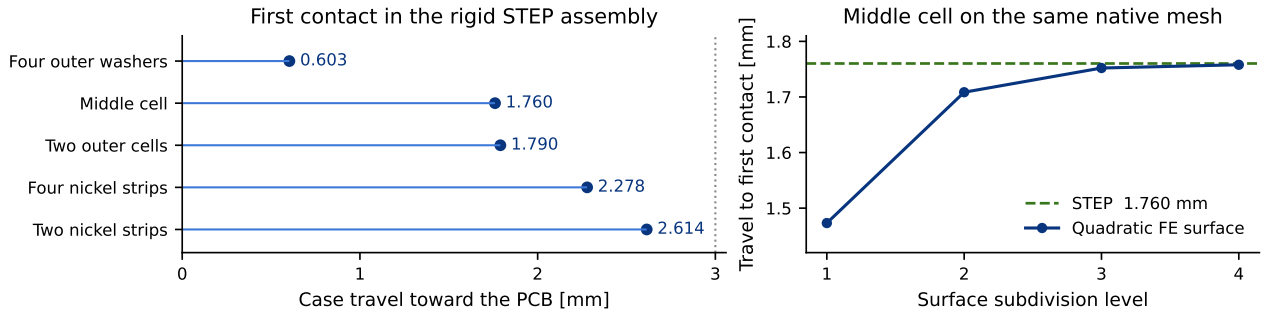


Figure 2: Left, downward travel from the 3 mm position. Solid intersections 0.05 mm beyond each onset independently confirm contact. Right, subdividing the same curved finite-element surfaces moves the middle-cell contact toward the STEP solution. The underlying case mesh and the physical geometry are fixed.

Materials and mounting representation

The supplied single-solid STEP has volume 63.069 cm^3. At a printed-PLA reference density of 1.17 g/cm^3, a dense case weighs 73.79 g, about 29.3% of the bare assembly mass. The material axes place the print layers parallel to the PCB for this scenario. PolyLite PLA coupon tensile moduli are 3.427 GPa in the print plane and 3.065 GPa through the layers. Shear moduli and Poisson ratios are engineering assumptions. Actual infill, orientation, print parameters and case mass require measurement. The published coupons used 100% infill. Polymaker technical data sheet.

The candidate model uses four M4 beam shanks with an effective tensile area of 8.78 mm^2. The comparison uses 69 GPa for aluminium and an exploratory 3.0 GPa for nylon. The nominal nylon hardware is McMaster-Carr 93640A145, M4 × 0.7 thread, 12 mm long. The represented bearing patches transfer loads between each end and its solid mesh. Compression-only members represent tab seating, with an assumed PLA annulus at the raised attachment points. The four case screws have no measured installation torque. The 750 N in the reference applies to the six original battery fasteners. McMaster-Carr metric socket-head screws.

Numerical checks and remaining comparisons

The initial mesh has 16,108 curved quadratic tetrahedra. A 43,659-element refinement is available. Both have positive minimum scaled Jacobians. The native preparation check counted four screw beams, four compression seats and 8,038 candidate case contact faces. The contact calculation must establish residual penetration, support reactions, seated shape and stresses before modal or random-response comparisons can support a performance claim. Both native bare-cell pulls first registered damage around 43 N with the assumed nickel law. A physical breaking load remains undetermined.

The bare-board repeat reproduces the six locked bolt loads

The original battery attachment provides the initial stress field for the connected modal and vibration analyses. In the first nonlinear load step each of the six pretension elements receives 750 N. In the second step MAPDL holds the attained section adjustment fixed. The reaction at the pretension degree of freedom then measures how much load the locked joint retains. The four case screws are absent from this reference run.

Existing bolt	Locked adjustment [μm]	Locked reaction [N]
1	28.297	749.99935
2	22.205	749.99945
3	18.597	749.99908
4	30.457	749.99680
5	20.686	749.99942
6	18.772	749.99868

Table 2: Fresh MAPDL extraction after length locking. Every row agrees with the October export at its stored precision. The largest departure from the applied 750 N is 0.00321 N.

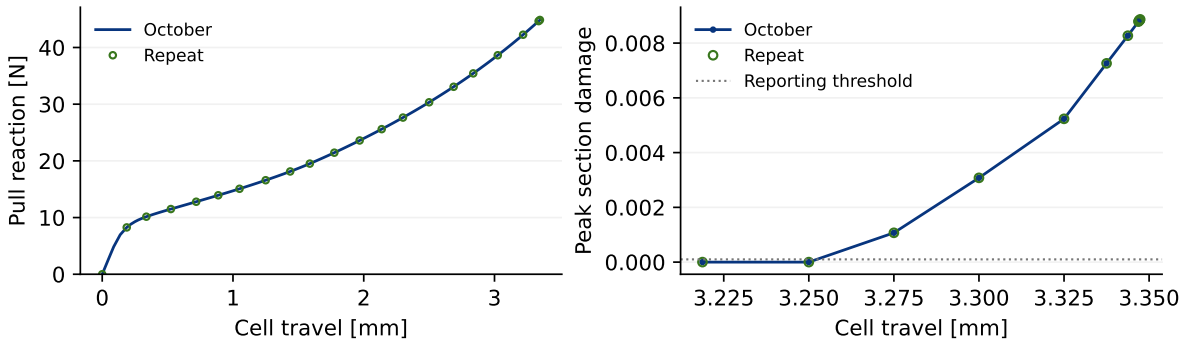
The solver stored eight equilibrium states and completed the two load steps in approximately 59 minutes on 12 distributed processes. The largest total displacement at lock was 0.0474595 mm. The repeated result differs from the October Mechanical export by 3.4×10^{-8} in relative terms. The model produced 45 warnings, including overlapping contacts, automatically removed redundant constraints and the fixed orientation of pretension elements during large-displacement analysis. The warning text printed in the main solver listing matched the reference run. This comparison establishes repeatability for the existing attachment. Page 7 compares the locked bare and raised assemblies. The vibration response requires separate modal and random-response results.

How the side-pull comparison measures load transfer

The local model drives the outer cell sideways through its rigid pilot while the two nickel strips bend. In the washer-clamped fixture, each strip is compressed with 750 N and the clamp height is then locked. During the pull, the cell can translate in the other two directions and rotate. The cased fixture adds a deformable printed enclosure, four screw shanks and compression-only tab seats. The neighboring cells and screw tops are fixed in this local approximation. The native model check counted 9,502 potential PLA contact faces. These contacts carry compression between PLA and the cell or washer surfaces. Direct PLA-to-nickel contact is omitted from this local fixture. That omission matters most in the flush geometry, where the CAD also intersects the strips.

The audit sums external forces and moments about the displaced cell centre at each converged increment. With the bilinear nickel law, the fresh bare fixture completed all 33 substeps. It required 8.542 N at 0.500 mm of travel, with peak equivalent plastic strain of 0.01254. Page 6 compares that fixture with both raised cases.

With the continuous nickel-damage law, the October run and its fresh repeat reached the same 110 equilibria. Both first crossed the 10^{-4} damage threshold between 3.250 and 3.275 mm, at 42.87–43.37 N. The maximum force difference over the two curves was 2.5×10^{-8} N. At their last converged state (3.347 mm, 44.86 N), four flat shell elements exceeded the threshold on the bottom face, two per strip, over 1.50 mm^2 of original shell area. The force was still rising when convergence stopped. The rupture load remains undetermined.



Native bare edge-cell pull through the last converged equilibrium. The right panel enlarges the onset interval. Damage is the largest reported shell-section value in the two strips.

Both default-contact meshes reproduce the October modal spectra

Each unpreloaded model used the archived mounting constraints and searched for 20 undamped natural frequencies between 1 and 10,000 Hz. The original mesh returned modes from 62.805 to 2205.547 Hz and the refined mesh from 66.611 to 2247.595 Hz. The maximum relative differences from the corresponding October exports were 1.78×10^{-8} and 1.74×10^{-9}. Repeating a given discretization thus produced the same frequency spectrum to the stated precision.

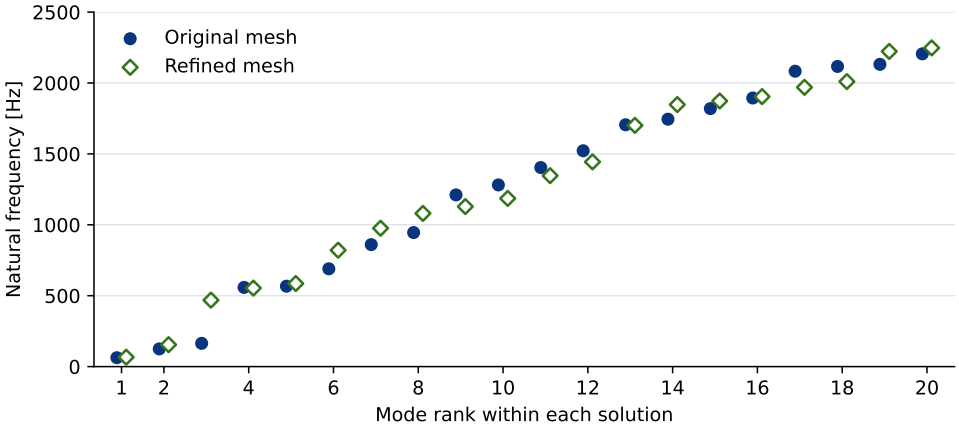


Figure 3: Fresh ANSYS results with the original default contacts. Rank orders the frequencies within each mesh. A shared rank can contain different mode shapes, particularly where modes exchange order.

The first original-mesh run returned only four modes and exited normally. The repeat kept the same input geometry, material properties, supports and search range. It changed the Block Lanczos algorithm to its alternate formulation and requested the Sturm sequence check. This recovered the complete spectrum. The default models retain the detached nine-body group in the earlier connection diagnosis. The bearing, locking and contact states of the six-bolt model provide the starting point for the connected vibration comparison.

PCB stress over the 480 frames of the seeded motion movie

Each of the three 9 s October base records has 589,825 samples at 65,536 Hz. After a 0.1 s ramp, the periodic 20–2000 Hz input reaches 14.136 g RMS. The bare-board native runs used 38 prestressed modes. The raised clips use independent integration on their native modes. All use 2% damping. Each clip shows 480 signed states from 1.000061 to 1.029297 s at 60 frames per second.

The six signed dynamic stress components were summed at each of 312,112 PCB element corners before calculating von Mises stress. The fresh native X history reproduced all 480 October bare-board frame maxima within 0.000028%; the independent integration agreed within 0.00182%. The largest value in this particular clip was 154.41 MPa bare, 169.94 MPa with raised nylon screws and 151.03 MPa with raised aluminium screws. The nylon peak occurred about 3.85 ms earlier and at a different board node from the bare peak.

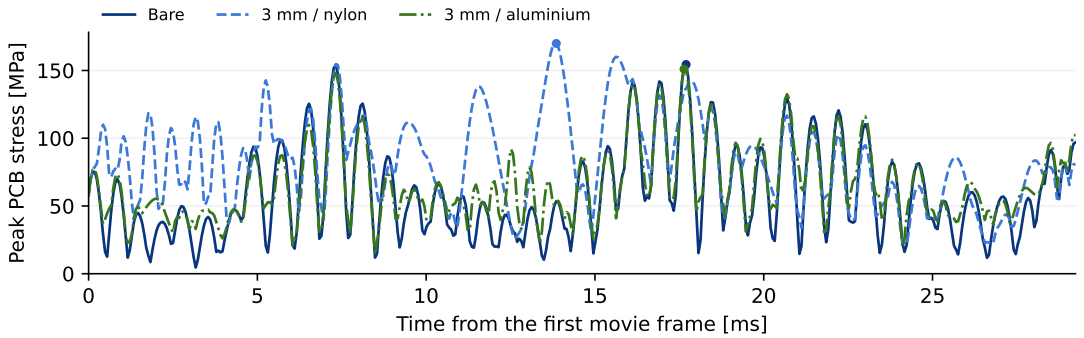


Figure 4: Maximum dynamic PCB stress at each of the original 480 movie times. The three curves use independent coordinates on each tightened modal basis. Each dot marks a clip maximum.

Seven seeded pairs share colour scales, 50-fold displacement magnification and 273-fold slow motion. All twelve camera orbits show native MAPDL one-sigma displacement fields for the bare board and both raised mounts. Paired movie catalogue.

Does the contact model start with the gap measured in CAD?

An enclosure that presses on the cells at the start of a raised-case calculation will appear to stiffen the assembly before the real parts would touch. We checked the undeformed contact pairs separately from the load solution. The local fixture uses one target for the moving edge cell and another for the fixed neighboring cells and washers. The global assembly uses deformable targets. The search radius tells MAPDL which nearby facets to examine; it does not set the physical clearance.

At 3 mm tab stand-off	Default search	0.6 mm search	2 mm search
Local: moving edge cell	+0.212	+0.212	+0.212
Local: fixed cells and washers	−7.685	+0.200	+0.200
Global: enclosure to assembly	+0.199	+0.199	+0.199

Table 3: Initial native contact check in millimetres. Positive values are gaps; negative values are overlaps. Default search radii ranged from 10.6 to 11.1 mm. The 1 mm search reproduced the 0.6 and 2 mm columns.

The CAD solids do not intersect at this stand-off. We rejected the first raised local calculation because its 7.685 mm overlap contradicted that independent check. It did not reach a converged load increment. The replacement uses a 1 mm search radius for the two added contact pairs and leaves the nickel clamp contacts unchanged. The comparison calls for the cell travel at first closure, contact pressure, penetration, and load transferred to the case mounts.

The finite-element gap is sampled at contact detection points on a curved mesh. The closest point on each exact cell surface is 0.144–0.161 mm from the case. The sideways travel to first rigid contact at the edge cell is 0.163 mm in positive X and 0.308 mm in negative X. Those are different measurements: the shortest distance can be oblique, and a loaded cell can rotate. Agreement among search radii checks the search procedure; it does not establish mesh convergence of contact onset or case stiffness.

Why the flush installation needs a separate check

The flush STEP has real interference. In the local model, a 2 mm search recovered the 1.178 mm crown overlap of the moving cell. A 1 mm search found only 1.000 mm, and a 0.6 mm search found only 0.598 mm. Tightening the search around the flush configuration can therefore hide real interference. Conversely, the default global search reported a peak overlap of 10.093 mm, versus 1.848 mm with a 2 mm search. These values identify a contact-model discrepancy. They do not establish an installation force. The flush comparison requires a verified path into contact, including the washers and strips, and force balance in the installed assembly before the vibration solve.

Both raised cases restrain the edge cell after contact

The local fixture pulls the outer cell in positive X. Its other translations and rotations are free. With the case on a 3 mm stand-off, the last open state was at 0.196 mm and the first closed state at 0.221 mm for either screw material. Under the 8.54 N load that moved the bare cell 0.50 mm, both cased cells travelled about 0.20 mm. The interpolated estimates were 0.207 mm with nylon and 0.202 mm with aluminium. Both lie in the same 0.025 mm interval between native states.

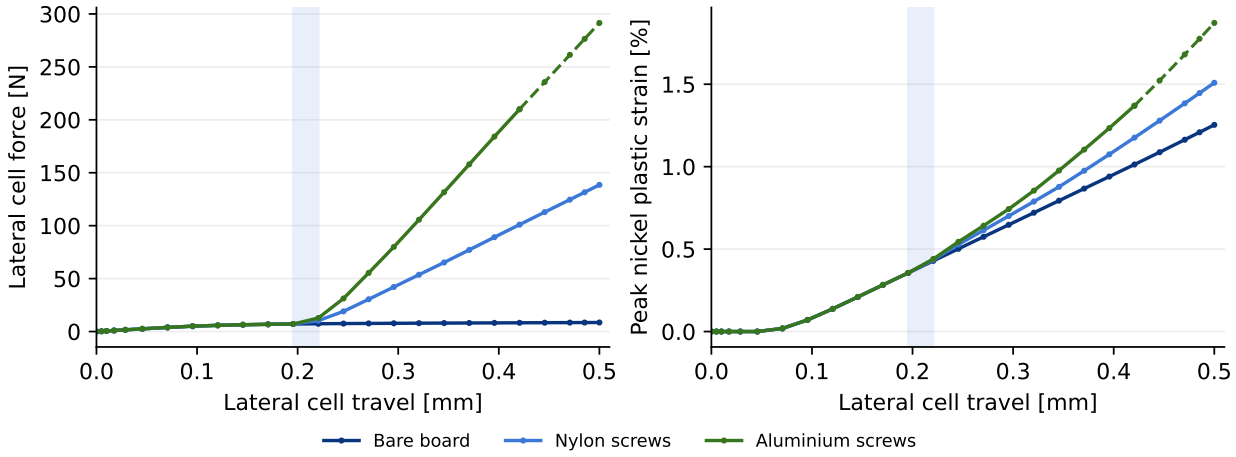


Figure 5: All three native pull histories. The blue band brackets first contact. The dashed aluminium segment extends from the last state below both PLA tensile references to states above at least one reference.

At 0.420625 mm prescribed cell travel	Bare	Raised nylon	Raised aluminium
Cell reaction in X [N]	8.26	100.99	209.96
Load on four case supports in X [N]	—	80.00	174.54
Peak equivalent plastic strain in nickel [%]	1.013	1.176	1.370
External work applied to cell [mJ]	2.63	11.91	22.56
Energy stored in printed case [mJ]	—	5.41	12.93
Energy stored in four screw shanks [mJ]	—	2.05	2.61

At a fixed travel the stiffer aluminium attachment increased both the case load and the strain in the nickel. At a fixed 8.54 N, the native states bracketing that force instead gave 0.36–0.44% peak nickel plastic strain for the two cased curves, versus 1.25% bare. Integrating native nodal forces and moments over the displacements of the PLA and shanks independently reproduced their stored energies within 0.009% of each component's maximum change.

Where the aluminium comparison leaves the coupon range

With aluminium screws, peak in-plane PLA tension crossed 52.3 MPa between 0.421 and 0.446 mm. Through-layer tension crossed 40.5 MPa between 0.446 and 0.471 mm. Both peaks were within 0.54 mm of the idealized screw bearing. At 0.50 mm the elastic continuation returned 291.5 N with aluminium, versus 138.5 N with nylon and 8.54 N bare. The nylon case stayed below both references, reaching 28.6 and 23.5 MPa. Its most tensile shank fibre reached 30.5 MPa, including 29.3 MPa from bending.

All fixtures saved 33 converged states. Peak penetration at a loaded cell/case corner was 0.0116 mm with nylon and 0.0198 mm with aluminium. The native cell contact forces and case-support reactions balanced to within 0.0019 N. These pulls assume fixed screw tops and omit direct PLA-to-nickel contact. At 0.421 mm cell travel, the axial-load resultants at the four tab bearings acted 5.41–6.54 mm from the screw axes with nylon and 5.41–5.49 mm with aluminium. All eight offsets exceed the modeled 3.75 mm bearing radius. A one-sided axial pressure across that footprint cannot carry the same force and moment. The computed pull stiffness therefore depends on how the installed joints transmit a couple. Page 17 gives the force check.

Both raised cases reduce board deflection during tightening

The bare board and both 3 mm stand-off assemblies completed eight equilibrium states under the same six-bolt tightening sequence. Each original bolt carried 750 N before its length was locked and retained 749.996–750.000 N afterward. The two cased models differ only in the properties assigned to the four case screws. The load in this comparison comes from the six original bolts.

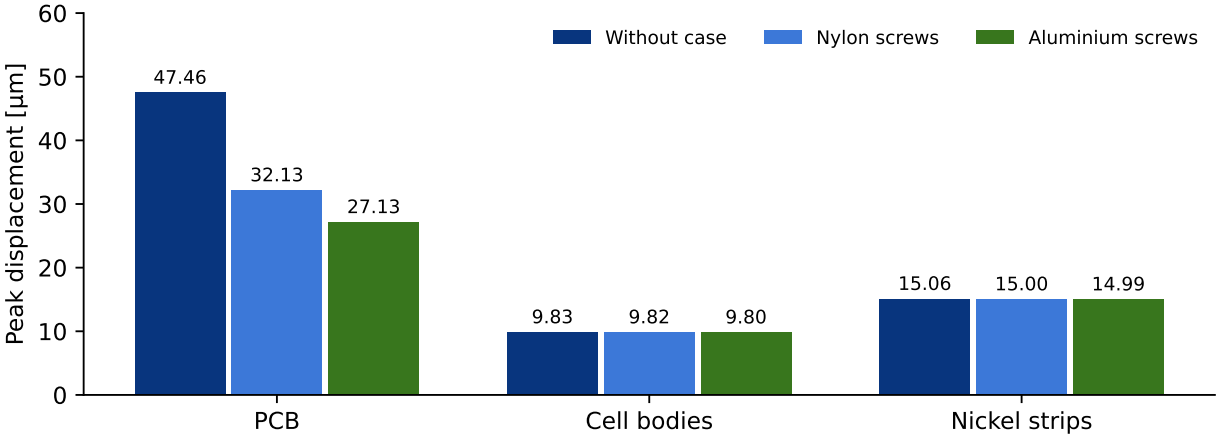


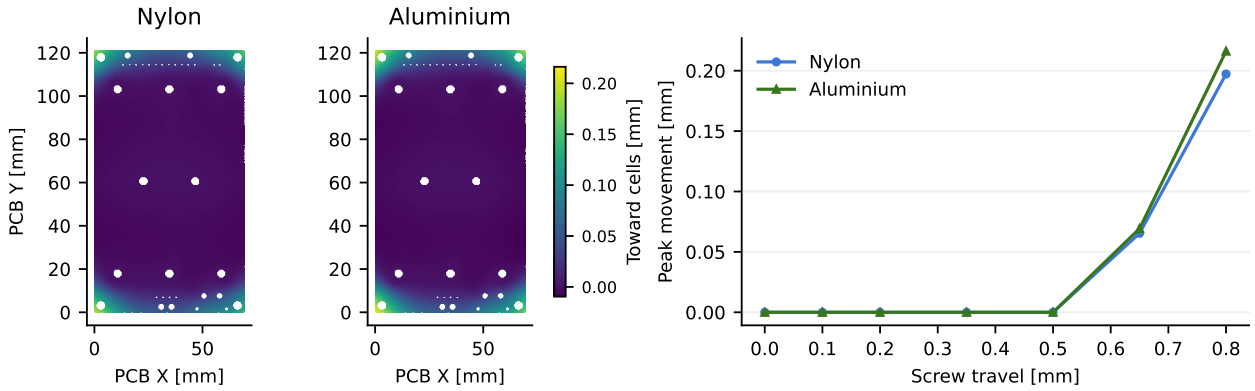
Figure 6: Peak displacement magnitude relative to the undeformed assembly after bolt locking. Each comparison uses the same mesh nodes on each part. Micrometres are used here to show the small changes at the cells and nickel strips.

The peak PCB displacement occurred at node 207485 in all three models. It fell from 47.46 μm bare to 32.13 μm with nylon screws and 27.13 μm with aluminium screws, reductions of 32.3% and 42.8%. Changing the four case screws from nylon to aluminium reduced this peak by another 5.00 μm, or 15.6% relative to nylon. The cell-body peak changed by at most 0.30% from bare and the nickel-strip peak by 0.51%. All 8,038 case-to-assembly contact faces stayed open in both raised assemblies, with zero contact pressure at every stored state. The case altered board bending through its four corner attachments.

ANSYS DPF recovered translations on 217,704 common PCB nodes. On the bare control the two displacement readers agreed on every component. The six pretension adjustments agreed with native MAPDL readback within 10^{-11} m in all three assemblies.

The PCB bends toward the cells in both flush installations

With either screw material the board moved less than 1 μm from its locked state through 0.50 mm of screw travel. At 0.80 mm, after the outer washers contacted the case, the same PCB corner at $(0, 0)$ had moved 0.197 mm toward the cells with nylon and 0.216 mm with aluminium. The aluminium screws increased the total drive force by 44% (page 19) and the peak board movement by 9.6%. The four PCB screw bearings moved 0.104–0.145 mm with nylon and 0.125–0.163 mm with aluminium. At 0.95 mm, the nylon model's corner movement reached 0.328 mm.



Upper PCB face at 0.80 mm screw travel, with one displacement scale for both maps. Positive movement is toward the cells. The curves show the maximum across all 217,704 board nodes relative to six-bolt lock. The two native meshes have identical PCB nodes and element connectivity.

The raised enclosure adds resonances below the first bare-board mode

We extracted undamped natural frequencies about the verified bolt-locked states, using the same mounting constraints and a 1–4000 Hz search. The assembled masses were 251.82 g bare, 325.82 g with the raised nylon-screw case and 326.10 g with aluminium screws. The models returned 38, 49 and 48 modes, respectively. The lowest three frequencies were 253, 289 and 339 Hz with nylon and 320, 372 and 378 Hz with aluminium. All six fall below the first bare-board mode at 486 Hz.

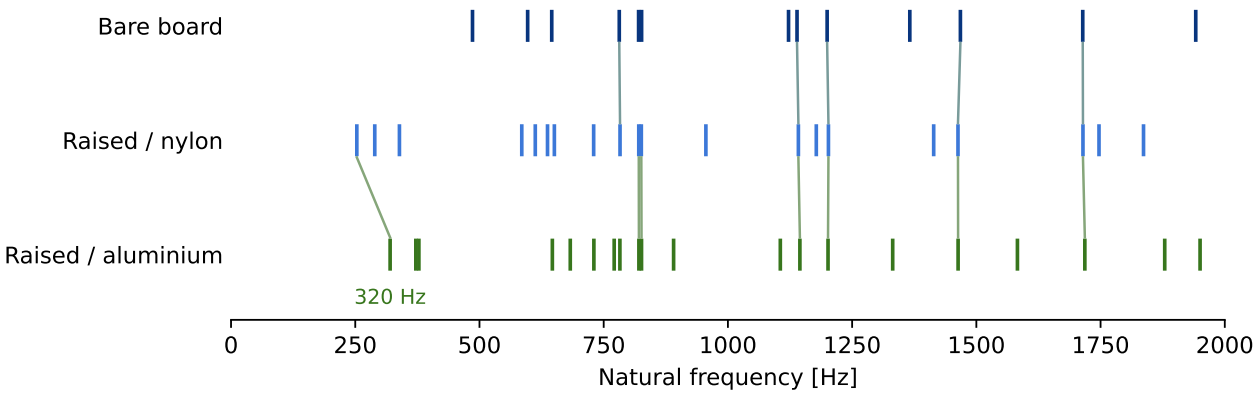


Figure 7: Each tick marks one native frequency in the excitation band. Lines connect five bare-to-nylon matches above MAC 0.94, and nylon-to-aluminium matches above MAC 0.97 where the assigned match is also the best available match.

The 253 Hz mode carried 17.9% of the assembled mass in X translation. In this shape the largest cell displacement was 2.8% of the peak on the PCB, while the peak on the case was 126% of the PCB peak. Those are amplitude ratios *within a mode*; the actual vibration displacement requires a forcing spectrum and damping. The 253 Hz mode had no close counterpart in the bare set: its highest shape correlation was 0.56. Conversely, the bare 781.35 Hz mode matched cased mode 9 at 782.88 Hz (correlation 0.943). This battery-motion mode shifted by only 0.20% even though its rank changed from fourth to ninth.

Effective translational mass, fraction of whole assembly	X	Y	Z
Bare, all 38 retained modes	87.2%	89.0%	89.6%
Raised nylon, all 49 retained modes	90.1%	91.5%	90.8%
Raised aluminium, all 48 retained modes	90.1%	91.4%	90.3%

How the shape comparison was made

Both modal meshes contain their assembly's static deformation. Before comparing vectors, we checked every original solid node against its own preload displacement and matched the 369,584 node IDs and their undeformed coordinates. The modal assurance criterion (MAC) is the squared, unweighted correlation of the global X, Y and Z eigenvector components at these nodes. It ranges from zero to one and is independent of the sign and normalization of a mode. It excludes the new case nodes so that both vectors describe the same structure. Modes with weak or competing matches remain ambiguous. For the two screw materials we also matched all 399,194 solid nodes, including 29,610 PLA nodes. The first mode rose by 26.5% from 253.16 to 320.17 Hz with full-solid MAC 0.989 and PLA-only MAC 0.994. It is a closely corresponding shape. The next two matches had full-solid MACs of 0.739 and 0.785.

The October bolted movie shows mode 20 at 2364.18 Hz. The two raised assemblies have mode 20 at 1836.45 and 1950.27 Hz. Its MAC with the bare mode is below 0.00023 in both. Searching the full native bases instead places the closest shape at mode 27, at 2365.06 Hz with nylon (MAC 0.717) and 2368.47 Hz with aluminium (MAC 0.640). The movie preserves the original rank and identifies this match on screen.

The linear model predicts more PCB motion with the raised case

We applied the October base acceleration spectrum separately along X, Y and Z to the three bolt-locked assemblies. For the 3 mm stand-off with nylon screws, the independent calculation predicts greater out-of-plane PCB motion for all three excitation directions. Under X excitation the peak PCB response in Z rises from 23.37 to 145.55 μm RMS, a factor of 6.23. In the same calculation the largest cell response in X falls from 61.43 to 53.79 μm RMS (12.4%). With aluminium screws, the peak PCB Z responses were 120.00, 82.33 and 100.55 μm RMS for base X, Y and Z. Compared with nylon, these fall by 17.6%, 5.0% and 23.6%. In particular, the 32% reduction in static board deflection during tightening did not carry over to this vibration metric.

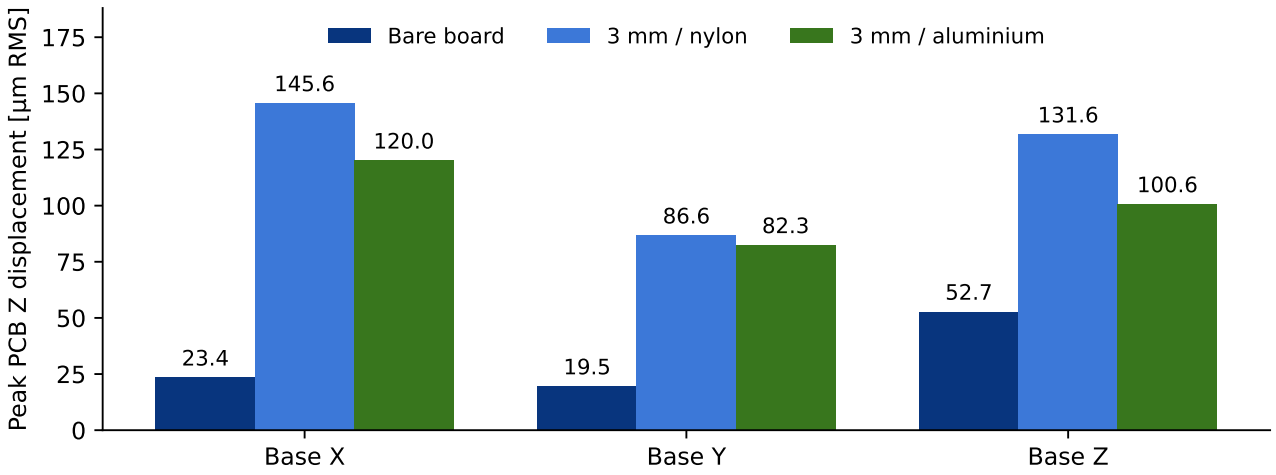


Figure 8: Each group compares the largest board-normal RMS displacement anywhere on the PCB under the same single-axis base spectrum. Peaks can occur at different nodes. Values describe motion relative to the supports.

How we checked the response

The independent calculation uses each native mode's signed displacement field and directional participation factor. We integrate the modal frequency responses against the one-sided acceleration spectrum (20–2000 Hz, 14.14 g RMS, 2% modal damping), including cross terms between modes driven by the same base input. This preserves cancellation and reinforcement. Halving the quadrature spacing from 0.25 to 0.125 Hz changed the modal covariance by less than 6×10^{-8} in relative norm. For the bare assembly we compared all 18 peak directional displacements: three base directions, three response components, and both whole-assembly and PCB scopes. The worst difference from October's native ANSYS random-response output was 0.013%. The three native MAPDL repeats reproduced all 51 October Mechanical maxima within 4.75×10^{-8} relative, including displacement, absolute acceleration, equivalent stress and the three normal elastic strains on the PCB. Peak PCB equivalent stresses under base X, Y and Z were 90.325, 17.850 and 32.482 MPa RMS.

We repeated the independent integration with modal cutoffs of 2000, 3000, 3500 and 4000 Hz, keeping the input spectrum fixed. For the raised nylon case under base X, both the 20-mode and 49-mode calculations gave a peak PCB Z displacement of 145.55 μm RMS. Across all three inputs, increasing the cutoff from 3500 to 4000 Hz changed the nine PCB directional peaks by at most 0.22% bare and 0.056% cased. The bare PCB's Y response to base Y differed by 8.32% with the 2000 Hz cutoff. These checks apply to displacement within the available basis.

The two native base-X solutions independently gave 120.00 μm for the raised aluminium case and 145.55 μm for the raised nylon assembly's peak PCB Z displacement. All 30 directional displacement peaks across five part selections per cased model agreed with the independent integrals within 0.0017%. Both calculations linearize about the tightened contact state.

The 3 mm stand-off does not prevent dynamic cell contact in the model

The tab stand-off is not the air gap at the cells. The smallest sampled cell clearance on the tightened mesh was 0.143 mm. We paired nearby opposing surfaces and calculated their simultaneous relative motion along the local separation direction. Under base X the most exposed cell location had 0.145 mm clearance and 0.098 mm RMS relative motion. The gap there is only 1.47 times that RMS displacement. This ratio suggests contact is plausible, but it does not tell us when contact occurs.

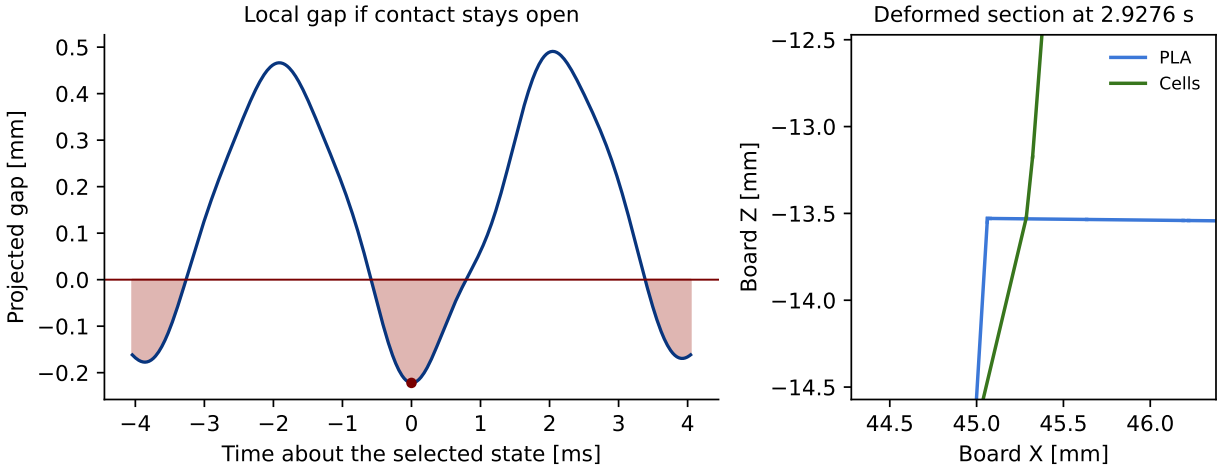


Figure 9: Raised case, nylon screws, October X-axis waveform. Left: first-order gap of the selected pair. A negative value indicates that the linear solution continues past projected contact. Right: actual-scale section of both triangulated surfaces displaced by the same signed modes. The curves are a diagnostic of the fixed-open model.

We independently integrated the nine-second October acceleration history and reconstructed the two skins at the instant of greatest projected approach for this pair, 2.9276 s into the waveform. At the tightened rest position the cell and case meshes had no intersections. In the selected deformed state, 895 triangle pairs intersected, and a separate signed-distance calculation placed 24 cell vertices more than 1 μm inside the PLA skin. The deepest sampled vertex lay 0.214 mm inside the PLA. Subdividing the curved elements twice more gave 0.215 mm at the same instant. Both surfaces had positive clearance before excitation. Their intersection shows where the fixed-open vibration assumption breaks down.

Excitation	Cell pair gap [mm]	Relative RMS [mm]	Intersecting cell/PLA pairs
Base X	0.145	0.098	895
Base Y	0.143	0.019	0
Base Z	0.236	0.112	0

Table 4: Each row selects the cell pair with the greatest RMS-to-gap ratio for that excitation, then checks the whole cell and case skins at one instant. A zero count at one instant does not establish separation throughout the record.

The surface search used both centroids and vertices of facing triangles within 1.2 mm in both directions, including the midside nodes of curved elements. Under base Z the nearest-line projection became negative, but the cell and case skins did not intersect at the selected instant. Tangential motion matters: a negative projected gap alone is insufficient evidence. Under base X both independent geometric checks found overlap. Until a nonlinear transient calculation or vibration test measures the response after closure, use the preceding fixed-contact RMS results as a comparison of the two linear models, not an installed flight stress, impact load or clearance margin.

The local flush installation stopped before the tabs seated

Four nylon screws drew the printed case toward a rigid board datum from an initial 3 mm stand-off. The existing washer clamps were tightened first. ANSYS saved 88 converged equilibria before the third load step failed at the next increment. The final converged screw-head travel was 2.8685 mm, leaving 0.1315 mm of the commanded travel. The solver never reached the sideways pull.

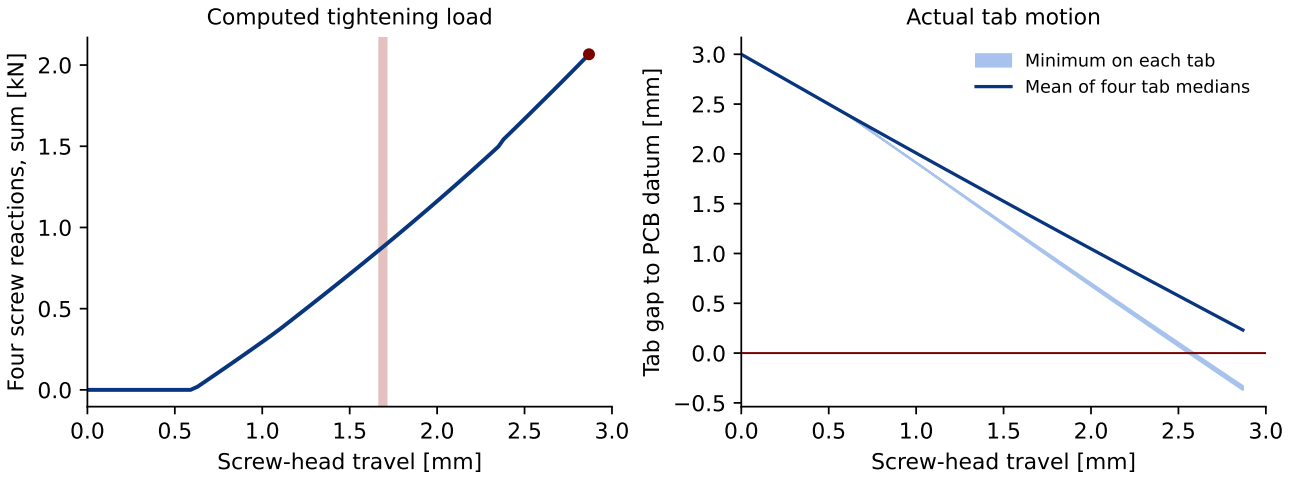
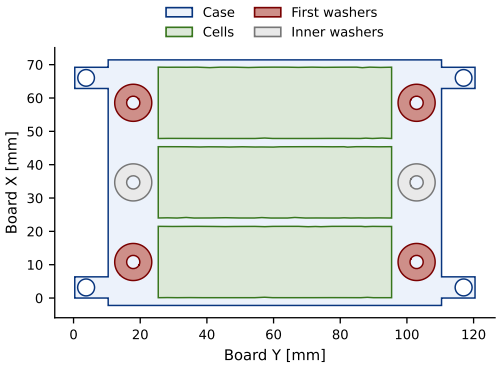


Figure 10: Left, the computed total screw reaction rose to 2.066 kN. The shaded interval marks the first exceedance of the 0.05 mm numerical penetration screen. Right, the tab surfaces tilted during drawdown. At the last equilibrium every tab crossed the nominal PCB plane at some points and remained clear at others.



The native fixture first loaded the four outer washers at 0.59–0.63 mm of screw travel. This brackets the independent STEP result of 0.603 mm (page 2). At 0.63 mm, two closed contact facets at each washer carried 4.89–4.92 N in board Z, totalling 19.63 N. Each loaded node lay within 0.8 μm of its assigned washer. The next nearest candidate was over 8.6 mm farther away. These local reactions apply to the fixture with fixed neighboring hardware.

The moving edge cell and its two nickel strips stayed clear throughout the partial path. Loaded-corner penetration first exceeded 0.05 mm at 1.67–1.71 mm travel and reached 0.119 mm. The largest departure from force balance was 0.030 N.

Final saved equilibrium	Tab 1	Tab 2	Tab 3	Tab 4
Screw-head reaction [N]	484	517	530	536
Minimum gap to board datum [mm]	−0.332	−0.357	−0.366	−0.369
Median sampled gap [mm]	0.248	0.227	0.236	0.214

In this local fixture the seating springs moved with the screw grips, allowing a tilted tab to cross the board plane. The peak computed PLA equivalent stress reached 233 MPa under the elastic print model. The physical installation force, failure sequence, and full-flush side-pull response remain unresolved. The whole-board installation model instead places contact directly between each tab and the flexible PCB surface, so the model can develop a normal reaction at the bearing land.

Contact deflects the raised case along the middle cell

The fixed-open calculation on page 10 placed parts of all three cell surfaces inside the enclosure at the selected X-waveform sample. Along the middle cell, the affected PLA facets extended from 25.2 to 95.6 mm in board Y. We therefore retained the whole length of this contact region in two native submodels. Their outer boundaries lay 9 or 15 mm from the middle-cell contact line. Each retained the original quadratic elements and directional PLA properties.

For each domain we imposed the tightened board displacements, then applied the signed motion at 2.9276 s to the two local cell bodies and the outer cut faces of the PLA. We increased the imposed motion from zero to its full value and unloaded in a series of static equilibria. With cell contact disabled, the free PLA nodes followed the parent solution to within 1 μm. With frictionless contact active, the largest change in their motion was 249 μm in the smaller domain and 244 μm in the larger one.

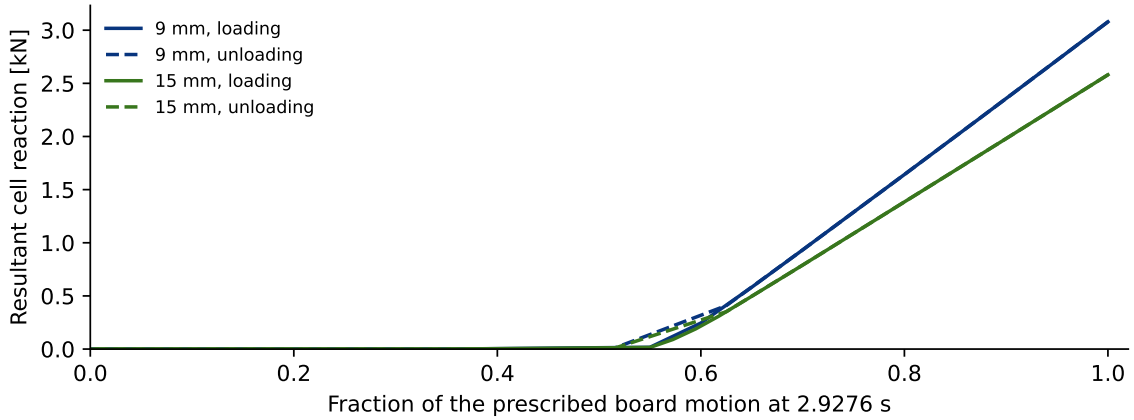


Figure 11: Each curve gives the force needed to impose this particular displacement path. The larger domain leaves a wider section of PLA free to flex. The peak reaction fell by 16.2%. The local models hold the cell motion fixed as the case pushes back.

At the fully imposed displacement	9 mm domain	15 mm domain
Resultant cell reaction [kN]	3.079	2.581
Active contact faces	24	24
Active faces touching the artificial cut	0	0
Maximum sampled skin overlap, contact disabled [mm]	0.2165	0.2164
Maximum sampled skin overlap, contact enabled [mm]	0.0424	0.0445

The independent geometry check measured signed distances on vertices and facet centres inside a common 8 mm band. Some tessellated surfaces still intersected even though the largest loaded-corner penetration recorded by ANSYS was below 0.5 μm at this state. Changing the cut size changed the force appreciably despite whole-model force imbalances below 0.004 N. These results establish local deflection under imposed motion. A flight impact load requires the cells and the surrounding board to respond to contact together.

A separate search near the beginning of the same X waveform found a clear middle-cell state at 85.3424 ms and intersecting skins at 85.3577 ms. The search sampled at 8192 Hz and refined that interval at 65,536 Hz. Contacts between earlier survey samples or elsewhere on the assembly remain possible. This interval sets the time window for the transient contact comparison.

The raised case increases PCB stress under Y and Z excitation

The native MAPDL comparison used the same X-axis base spectrum and 2% damping on both assemblies. Each original board bolt had already been tightened and locked. With four nylon case screws at the 3 mm stand-off, the peak PCB equivalent stress fell by 1.6%, from 90.325 to 88.847 MPa RMS. Its location moved about 54 mm across the board. The PCB's maximum normal strains in X and Y both increased.

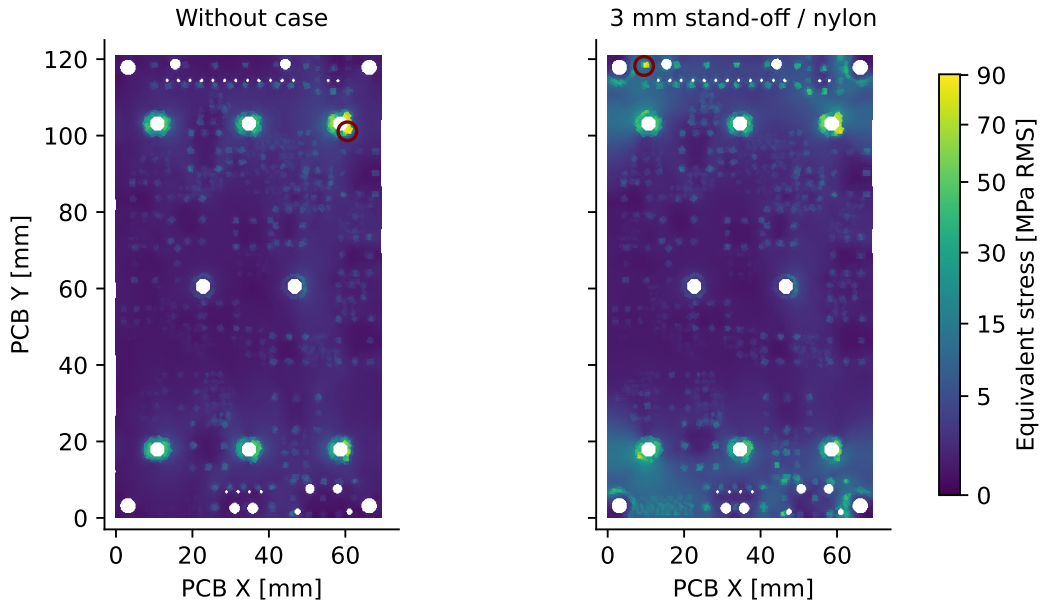


Figure 12: PCB plan view in the original mesh coordinates. Each solid footprint shows its largest stored equivalent-stress RMS among its corners. Higher values are drawn last through the thickness. Red circles mark the maxima at $(60.55, 101.11)$ and $(9.54, 118.21)$ mm. Both maps use the same colour scale.

Peak response, one sigma	Base axis	Bare	3 mm nylon	3 mm aluminium
PCB Z displacement [μm]	X	23.37	145.55	120.00
	Y	19.45	86.65	82.33
	Z	52.67	131.64	100.56
PCB equivalent stress [MPa]	X	90.32	88.85	88.65
	Y	17.85	57.05	50.29
	Z	32.48	64.77	84.19
Nickel equivalent stress [MPa]	X	291.90	256.24	286.46
	Y	15.11	21.85	21.86
	Z	38.13	38.38	38.20

Under base Z, aluminium screws gave less board travel than nylon screws and higher peak PCB stress. Each entry is a maximum over the named part. The spectra hold the tightened contact state fixed (pages 10 and 12).

An independent sum of the signed modal stresses reproduced all nine native stress-peak locations across 312,112 PCB element corners. Reducing the modal cutoff from 4 to 3.5 kHz changed the nine peak values by at most 0.20%. With only the modes below 2 kHz, the bare base-Y peak was 34.65% lower. The excitation band remained 20–2000 Hz throughout.

The printed tab carries a local load across its layers

The printed case and PCB react differently to the choice of screw. At the 3 mm stand-off, changing nylon screws to aluminium reduced peak PCB movement during tightening (page 7), while the greatest tensile stress normal to the layers anywhere in the PLA rose from 0.688 to 1.515 MPa. The independent orthotropic elastic-law check reproduced the stored tightening strains to within 4.0×10^{-8} relative field norm across all 64,432 element corners.

For X-axis vibration with nylon screws, the native peak layer-normal stress was 17.16 MPa RMS, 0.53 mm from the nearest node tied to a screw bearing. Among corners at least 2 mm from those nodes it was 5.72 MPa RMS. The static tension at the corner of greatest RMS was 0.688 MPa. The six native component fields and stored equivalent-stress RMS agreed with integration of the signed modal stresses to within 0.001% relative field norm.

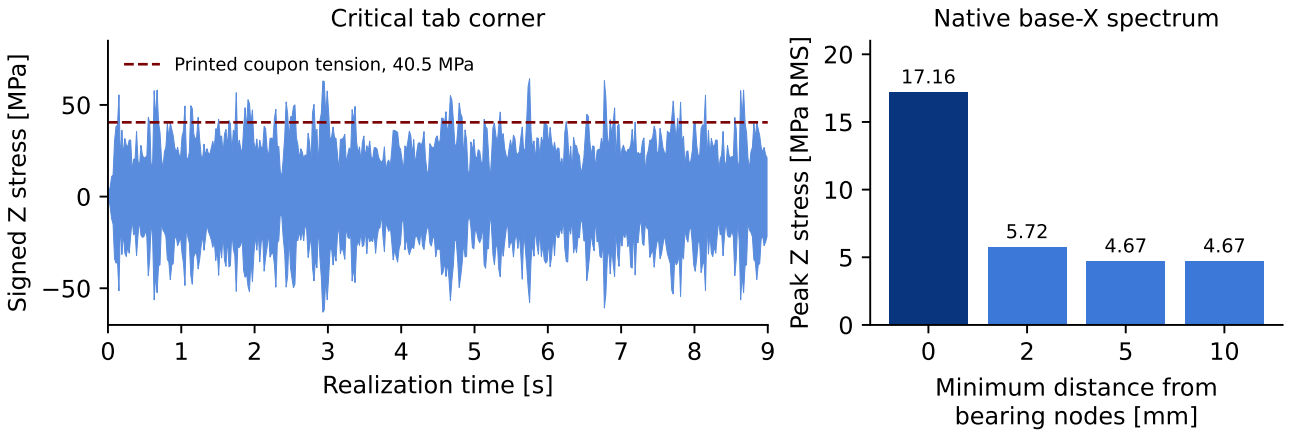


Figure 13: Left, independent reconstruction at element 1215221, node 815413, with the static tension included. The band preserves sample maxima and minima in 20 ms windows. Right, maximum stored through-layer RMS over locations at or beyond each distance from a coupled screw-bearing node.

At that corner the nine-second October X waveform gave a conditional maximum of 64.3 MPa in tension at 5.741 s. Its last eight seconds had a dynamic RMS of 17.15 MPa. For scale, the PolyLite PLA data sheet reports mean layer-normal tensile strength of 40.5 MPa on printed coupons. The assembly filament, print orientation, infill and screw seating have yet to be measured. The stress peak is close to an idealized rigid bearing. At that very sample, the cell and PLA surfaces intersect in 944 triangle pairs, with a maximum sampled depth of 0.190 mm. A conclusion about tab cracking therefore requires the bearing and post-contact response to be resolved, with strength data for the actual print.

Aluminium screws reduce board motion and concentrate load in the tab

The two raised models differ only in the properties of the four case screws. Both native base-X spectra used the same excitation, printed case and board mesh. Relative to nylon, aluminium reduced the peak PCB Z displacement by 17.6% and the PCB Z acceleration by 18.6%. The two PCB equivalent-stress peaks were within 0.22% of each other. Changing to aluminium moved the peak back to the same node as the bare reference.

Native base-X response, RMS	Bare	Raised nylon	Raised aluminium
PCB Z displacement [μm]	23.37	145.55	120.00
PCB Z acceleration [g]	52.29	67.45	54.92
Cell-body X acceleration [g]	103.51	91.81	100.90
PCB equivalent stress [MPa]	90.325	88.847	88.655
PLA layer-normal stress [MPa]	—	17.16	24.78
PLA layer-normal stress, at least 2 mm from bearings [MPa]	—	5.72	5.41

The aluminium case had a peak layer-normal stress of 24.78 MPa RMS at a corner 0.50 mm from a coupled bearing node. That is 44.4% above the nylon peak. Among corners at least 2 mm away, aluminium gave a slightly lower peak. This dependence on the bearing region motivates the finer case mesh and inspection of how the actual screw head bears on the print.

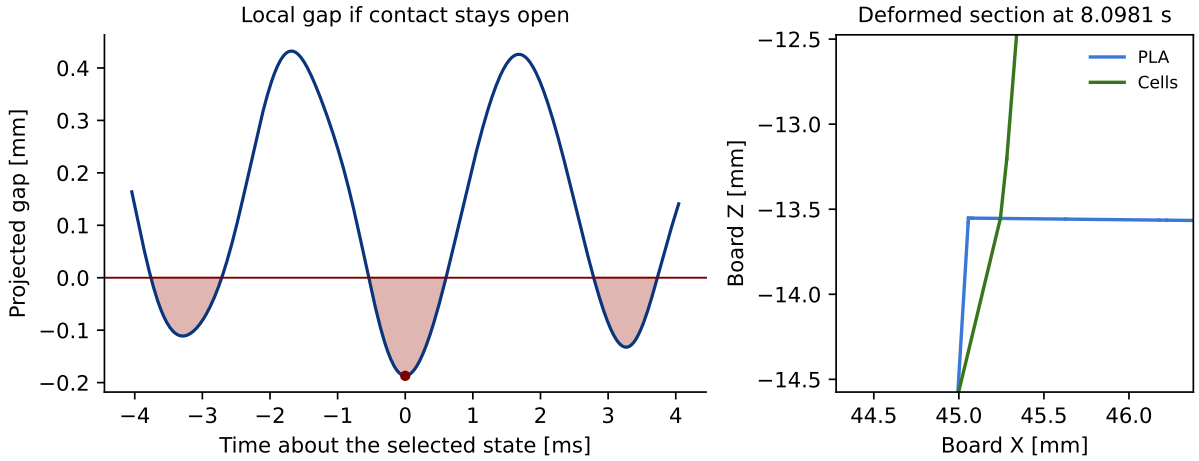


Figure 14: Raised aluminium-screw case, independent response to the October X waveform. Near the selected point, the tightened gap was 0.145 mm and the normal relative motion was 0.070 mm RMS, compared with 0.098 mm for nylon. The section is plotted at physical scale.

At 8.098053 s, the aluminium model's cell and PLA skins intersected in 864 triangle pairs. Nineteen sampled cell vertices lay more than 1 μm inside the case, and the deepest vertex was 0.174 mm inside. At 8.098083 s the independent signed reconstruction reached 110.1 MPa in layer-normal tension at the critical tab corner. At this exact sample the skins intersect in 854 triangle pairs with a deepest sampled cell vertex of 0.174 mm. For both screw materials, the peak tension at the selected highest-RMS corner occurs while the cells overlap the PLA in the linear solution. With two additional geometric subdivisions, the cell vertices reached 0.201 mm inside the PLA with nylon and 0.175 mm with aluminium at their respective peak-tension samples. The physical loads after contact and at the actual screw bearing remain unresolved.

The raised case lowers the highest nickel-strip stress under base X

Each cell connects to the board through a nickel strip at each end. We compared the same 11,348 element corners across all three assemblies under the October X-axis vibration spectrum. With nylon case screws, peak equivalent stress fell in all six strips. The largest value occurred at the same corner on the high-Y strip of the right-hand cell in all three assemblies. It fell from 291.9 MPa RMS bare to 256.2 MPa with nylon (12.2%) and 286.5 MPa with aluminium (1.9%). Aluminium raised the middle cell's two strip peaks by 2.6% and 2.7% relative to bare.

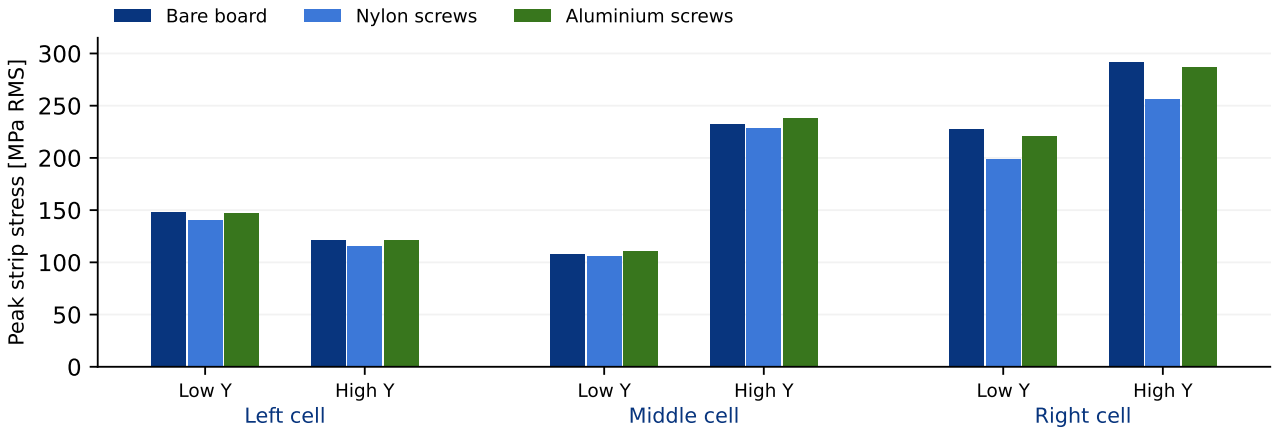


Figure 15: Native one-sigma equivalent stress, maximized separately in each nickel strip. Low and high Y identify the strip ends near board coordinates 22 and 99 mm. Left, middle and right refer to increasing board X. For every strip, its maximum stayed at the same element corner across screw choices.

The assumed nickel yield stress is reached while the cells still clear the case

The global model uses linear elastic nickel with modulus 220 GPa. The separate October nonlinear pull model assumes a 185 MPa initial yield stress for annealed nickel and a modulus of 205 GPa. To compare the two, we reconstructed the global *signed* stress history during the first 0.1 s of the seeded October base-X waveform. At each sampled instant we added the six stress components from the locked preload to their respective dynamic components, then calculated von Mises stress at each corner. Independent spectral integration also reproduced all six native RMS component fields and the stored equivalent-stress field to within 0.0033% relative field norm.

First sampled crossing of 185 MPa	Bare	Raised / nylon	Raised / aluminium
Base X, time bracket [ms]	38.116–38.132	38.956–38.971	38.116–38.132
Peak after tightening alone [MPa]	161.9	162.1	162.1
Base X, minimum clearance bound [mm]	—	0.038	0.068

Table 5: Each time refers to this excitation with its prescribed 0.1 s ramp. The material threshold is a property assumed in the local model. Actual strip temper and weld properties would be needed to infer yielding or fatigue life.

The clearance check covers every quadratic exterior face on all three cells and the PLA. At the tightened position their separation exceeded 0.130 mm. We bounded the subsequent motion of each face by tracking its quadratic Bézier control points in a common translating frame at 65,536 samples/s. The two bodies' maximum excursions summed to 0.092 mm with nylon and 0.062 mm with aluminium through their respective threshold crossings. This places the early stress excursion before direct cell-to-case contact in both calculations. The air gap can close later in the same waveform (pages 10 and 15).

With base Z excitation, only the raised aluminium model crossed 185 MPa during the first 100 ms. The first crossing occurred at 77.911–77.927 ms on the middle cell attachment (STEP body 17). The same moving-surface bound gave at least 0.0256 mm of cell-to-case clearance through that sample. With base Y, all three models stayed below the reference throughout the 100 ms window.

The local flush fixture exceeds the printed-PLA tensile references early in tightening

The four outer washers first loaded the nylon-screw fixture at 0.59–0.63 mm of screw-head travel. Between 0.91 and 0.95 mm, the greatest tensile stress in the assumed print plane crossed 52.3 MPa. Through-layer tension crossed 40.5 MPa between 0.99 and 1.03 mm. These are typical tensile strengths for the 100%-infill reference coupons. Comparing the native true stresses with these nominal coupon strengths gives an approximate check on the range of the elastic model. Polymaker coupon data.

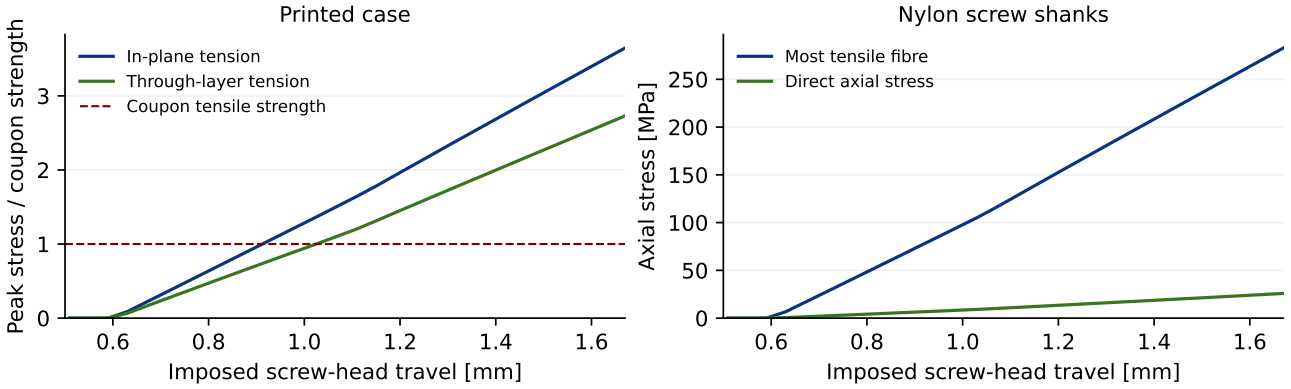


Figure 16: Native converged local states with rigid neighboring hardware and prescribed screw-head travel. Left, maximum tensile stress over the 16,108 quadratic PLA elements. Right, section-node tension including bending and the direct axial component in the four idealized short screw shanks. Curves stop at the last sampled travel before loaded penetration first exceeded 0.05 mm.

Local material reference	First corner crossing [mm]	First element with all four corners above [mm]
In-plane tension, 52.3 MPa	0.91–0.95	1.55–1.59
Through-layer tension, 40.5 MPa	0.99–1.03	2.57–2.61
Through-layer elongation, 1.8%	1.23–1.27	No sampled element

Table 6: The coupon elongation corresponds to logarithmic strain $\ln(1.018) = 0.01784$. The first in-plane and through-layer stress crossings occurred 3.72 and 3.78 mm from the nearest coupled bearing node. Contact penetration already exceeded the local screen by the time all four corners of an element crossed the through-layer stress reference.

At 0.95 mm travel each prescribed screw head carried 64.0–64.5 N along board Z. The largest direct axial screw stress was 7.37 MPa. Including bending, the most tensile modelled screw fibre reached 85.34 MPa. A rigid connection between the screw rotation and its bearing footprint strongly influences that value. The McMaster-Carr listing for the representative nylon screw (93640A145) gives its dimensions and lists tensile strength as unrated. Its installed strength, head rocking and thread engagement therefore remain unverified. McMaster-Carr screw data.

How much moment each tab bearing carries

At 0.91 mm of screw travel, before the first coupon crossing, the four local tab bearings carried 56.5–56.9 N each along board Z. For each bearing, we summed the axial nodal forces $F_{z,i}$ and their first moment about the screw axis. The equivalent offset $e = \|\sum_i \mathbf{r}_{xy,i} F_{z,i}\| / |\sum_i F_{z,i}|$ was 4.44–4.46 mm. Pressure acting in one direction inside a circular footprint of radius R gives $e \leq R$. For the modeled $R = 3.75$ mm, reproducing the observed moment using axial forces alone would require at least 5.2–5.4 N in the opposing direction per bearing. Forces from the printed solid and the screw/seat assembly agreed within 0.000003 N and moments within 0.00001 N mm. An actual joint can also transfer moment through the shank, thread or head restraint.

For comparison, tightening the six original bolts in the raised full-board models produced a maximum case-screw fibre tension of 1.15 MPa with nylon and 1.63 MPa with aluminium. Neither raised case reached a printed-PLA coupon reference during tightening. The local flush trial subsequently crossed the board datum (page 11). Resolving the force required for an actual flush seat depends on the full-board contact path and the as-built material and joint behaviour.

Does the cell touch the case during the motion shown in the movies?

The case with nylon screws lost its modeled cell clearance in 18 consecutive frames under X-axis excitation. Using the October numbering (first frame 1), these are frames 253–270 of the 480-frame film, about 4.20–4.48 s into playback and 1.015442–1.016479 s into the physical load history. At their midpoint, a cell surface vertex lay 14.6 μm inside the modeled PLA. The movie magnifies displacements 50 times. The section below uses the physical deformations.

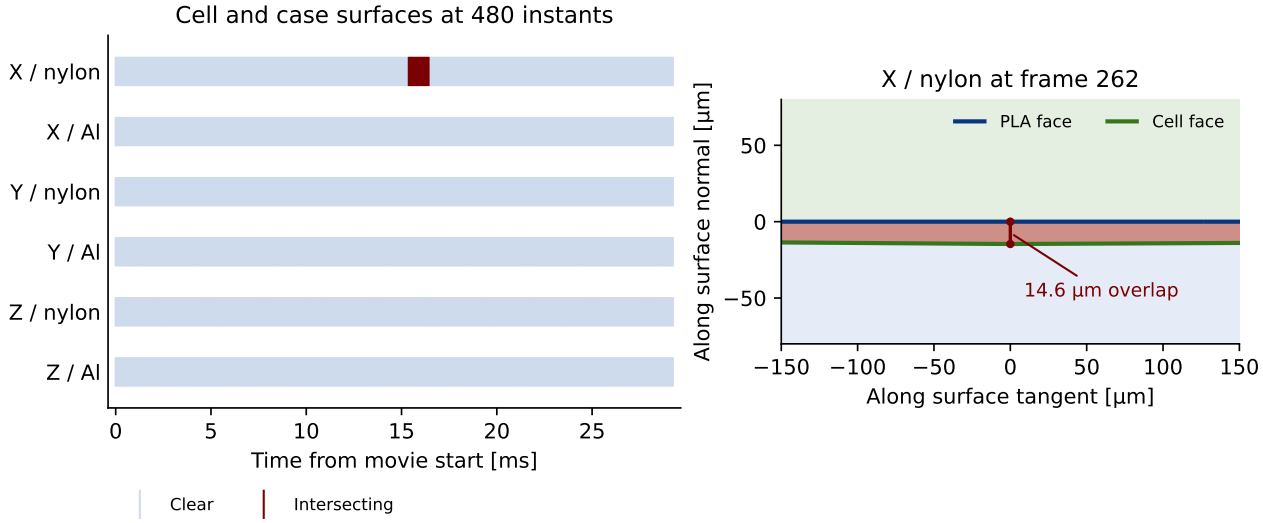


Figure 17: Left, the cell and PLA surface check repeated at every displayed instant for each 3 mm mount and input axis. Right, the exterior surfaces at frame 262, cut in a plane through the closest PLA point and the most deeply embedded sampled cell vertex. Both section axes use the same scale.

Raised mount	X	Y	Z
Nylon screws	18/480	0/480	0/480
Aluminium screws	0/480	0/480	0/480

Table 7: Frames with intersecting surfaces under separate base excitations. In the longer nine-second X history both screw variants showed interference at other sampled instants (pages 10 and 15).

At each mount's own peak dynamic PCB stress in the X clip, the smallest sampled cell-vertex gap was 0.149 mm at frame 228 for nylon and 0.146 mm at frame 290 for aluminium. The fixed-contact motion calculation omits any change in stress or trajectory caused by earlier contact.

How the surface check was verified

Independent modal coordinates displaced all 8,038 exterior quadratic PLA faces and all 938 cell faces. Each quadratic face lies inside the convex hull of six control points. We excluded pairs with disjoint hulls and subdivided overlapping hulls until they separated or successive triangular approximations intersected. In all 18 crossing frames the eight- and sixteen-segment edge refinements agreed. The remaining 2,862 frame/configuration combinations had disjoint curved surfaces.

An independent check using the full closed skins gave a sampled overlap of 15.0 μm at four segments per edge and 14.6 μm at eight for frame 262. The geometry check also reproduced a prescribed planar crossing and the earlier open and intersecting whole-assembly states. Impact force and the motion between displayed samples require further analysis. Raised cases under X-axis excitation.

At 2.75 mm only the nylon-mounted tabs touch the PCB

At equal screw travel, the native contact field showed edge contact on all four nylon-mounted tabs and zero contact pressure on the aluminium-mounted tabs. The surface reconstruction put the closest aluminium tab 0.204 mm from the board. In both models the four outer washers intersected the PLA, and all three cells remained clear. The smallest sampled cell gaps were 0.077 mm with nylon and 0.076 mm with aluminium. Both trials had already exceeded the printed-PLA tensile references and the hardware penetration limit. These late states describe the elastic models only.

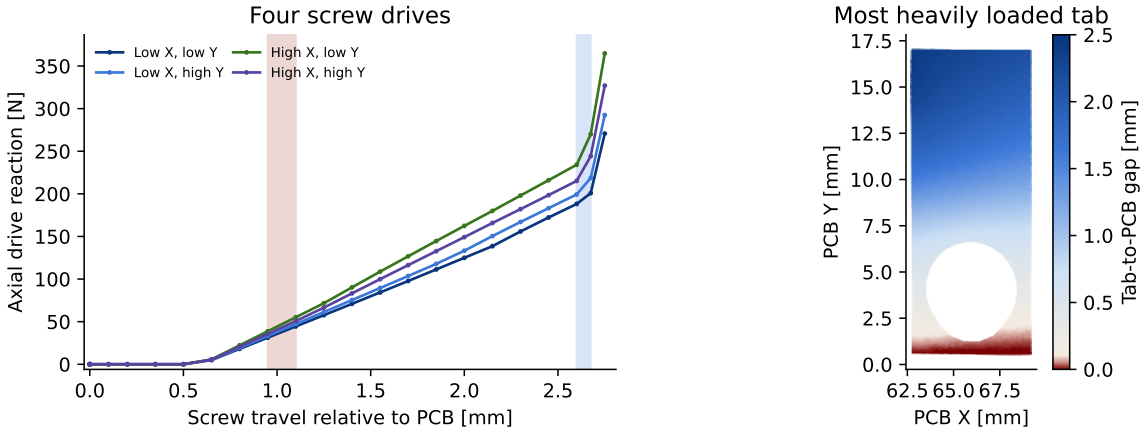


Figure 18: Nylon mount. Left, the four screw reactions. Red brackets the first through-layer coupon crossing. Blue brackets first loaded tab-to-PCB contact. Right, the deformed footprint of the most heavily loaded tab at 2.75 mm. Dark blue denotes the largest gap. The uncoloured bore has no opposing faces.

Tab position	Nylon near-seat area [%]	Nylon largest gap [mm]	Al smallest gap [mm]
Low X, low Y	4.4	2.425	0.218
Low X, high Y	4.7	2.422	0.217
High X, low Y	5.1	2.417	0.204
High X, high Y	4.9	2.418	0.210

Table 8: Near-seat area is the fraction of each overlapping tab footprint within 0.05 mm of the board. It was zero at every aluminium-mounted tab. All quantities are at 2.75 mm screw travel.

With nylon, the first through-layer tensile stress crossed 40.5 MPa between 0.95 and 1.10 mm, at a node tied to the idealized screw bearing. In-plane tension crossed 52.3 MPa at 1.10–1.25 mm. When the search excluded points within 2 mm of tied bearing nodes, both references were first crossed at 1.25–1.40 mm. The latter interval follows the first crossing of the 15 μm hardware-penetration screen (1.10–1.25 mm). At 2.75 mm the contact field reached 71 μm penetration. The nylon trial stopped at this equilibrium after three cutbacks. An installation-load rating requires measured print and contact behaviour.

The combined nylon result reproduced all 28 equilibria. The hardware contact audit included all 8,038 PLA contact faces. At 0.95 mm travel, before the first coupon crossing, the four original outer battery bolts gained 10.3–14.0 N above their 750 N locked preload. The two centre bolts changed by −0.9 and +1.4 N. At 0.95 mm the four drives supplied 25.59 mJ of work (28.41 mJ using alternate saved states). The PCB gained 8.81 mJ of stiffness energy, the PLA 8.01 mJ and the four screws 6.59 mJ. All element groups together accounted for 24.59 mJ.

Aluminium screws increase the load at the same drawdown

In the two whole-board trials the screw shank geometry was identical. Only modulus, Poisson ratio and density differed between the decks. The first loaded hardware contact fell between 0.50 and 0.65 mm of relative travel for both mounts. At 0.80 mm, the four drives together carried 80.0 N with nylon and 115.6 N with aluminium. Peak through-layer PLA tension was 18.2 and 37.2 MPa respectively, against the 40.5 MPa coupon reference. With aluminium, both the in-plane and through-layer tensile references were first exceeded between 0.80 and 0.95 mm, while all four tab contacts remained open. In the 0.80 mm aluminium snapshot only the four outer washers intersected the PLA. Refining the curved surfaces gave up to 24.0 μm of sampled washer overlap, compared with 7.9 μm at loaded native contact corners. The smallest sampled cell clearance was 0.066 mm. These early reactions depend on how the washer contact is represented.

The raised nylon case lowers the transverse load on all three cells

Under base-X vibration, the X component of the right-hand cell's net dynamic force fell from 30.51 N RMS bare to 26.67 N with nylon screws and 29.73 N with aluminium. These are reductions of 12.6% and 2.6%. Nylon also reduced this load on the left and centre cells. Aluminium raised the centre-cell load from 25.60 to 26.28 N RMS. The PSD calculation keeps the tightened cell-to-case gap open. The two nickel attachments on each cell therefore carry its net dynamic load until additional contact occurs.

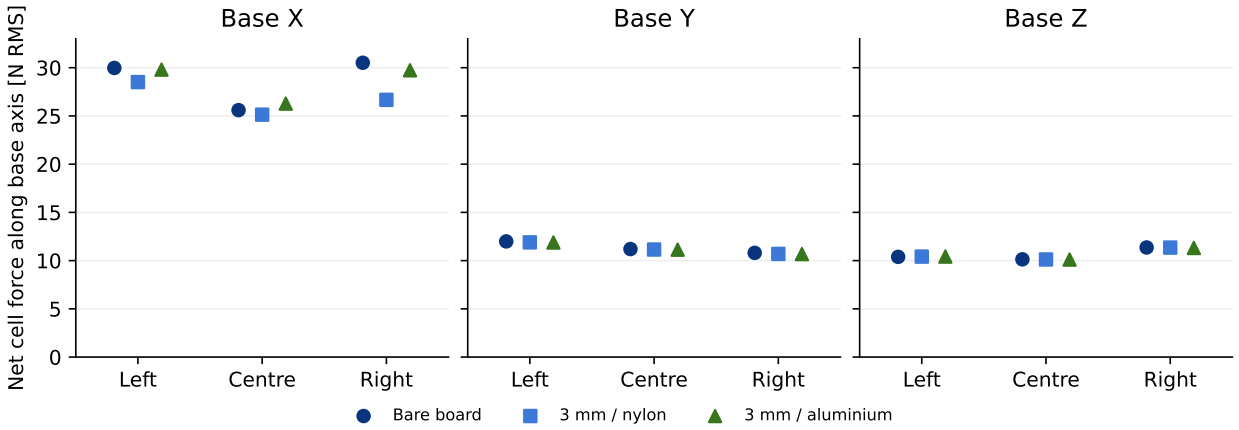


Figure 19: Net force component parallel to each of three separate base excitations. Markers within each group correspond to the same physical cell. The ordinate is shared across panels. The material model gives each cell a mass of 55.38 g.

Base X, cell force [N RMS]	Bare board	3 mm / nylon	3 mm / aluminium
Left	29.97	28.51	29.80
Centre	25.60	25.13	26.28
Right	30.51	26.67	29.73

Along the excited direction, the Y and Z cases changed by less than 0.13 N on each cell. A transverse component can still grow. With Y input the X force on the left cell was 0.12 N RMS bare, 1.75 N with nylon and 1.01 N with aluminium. Page 22 compares the signed forces at both cell-to-nickel attachments.

Mass averaging and verification

For each native mode we integrated the signed translations over each curved ten-node cell tetrahedron with its material density. The relative acceleration transfer function follows from differentiating modal displacement twice. We added the common base acceleration before integrating over the 20–2000 Hz spectrum. Combining the signed quantities in this order accounts for phase cancellation within the cell. Multiplying the resulting centre-of-mass acceleration by cell mass gives the force.

The four- and five-point-per-coordinate volume rules agreed within 10^{-9} in relative norm. Reducing the upper modal cutoff from 4 to 3.5 kHz changed the driven-axis forces by at most 0.30%. Independent nodal acceleration RMS fields matched all nine native PSD fields within 0.044% in relative norm over cell nodes. Separately, the original bare-board native time histories and original signed shapes reproduced the independently integrated seeded cell accelerations within 0.0075%.

For the eight stationary seconds of the historical forcing, the driven-axis RMS loads differed from the continuous-spectrum estimates by at most 0.88%. The largest discrepancy over all components was 10.4% and occurred in a smaller transverse component. Evaluating the same sampled waveform with the discrete Newmark transfer function reproduced the time integration. At one-quarter of the historical time step, its largest discrepancy from the continuous-frequency response at the same Fourier bins fell to 0.59%.

The raised case preserves the first unpreloaded mode

The first default-contact mode remained at 66.61 Hz for the bare board and both raised cases. The signed displacements of the 368,754 original solid nodes give a modal assurance criterion (MAC) above 0.999999 for both screw materials. The 20th modes tell a different story. Although each is the 20th frequency in its own assembly, the three shapes differ substantially.

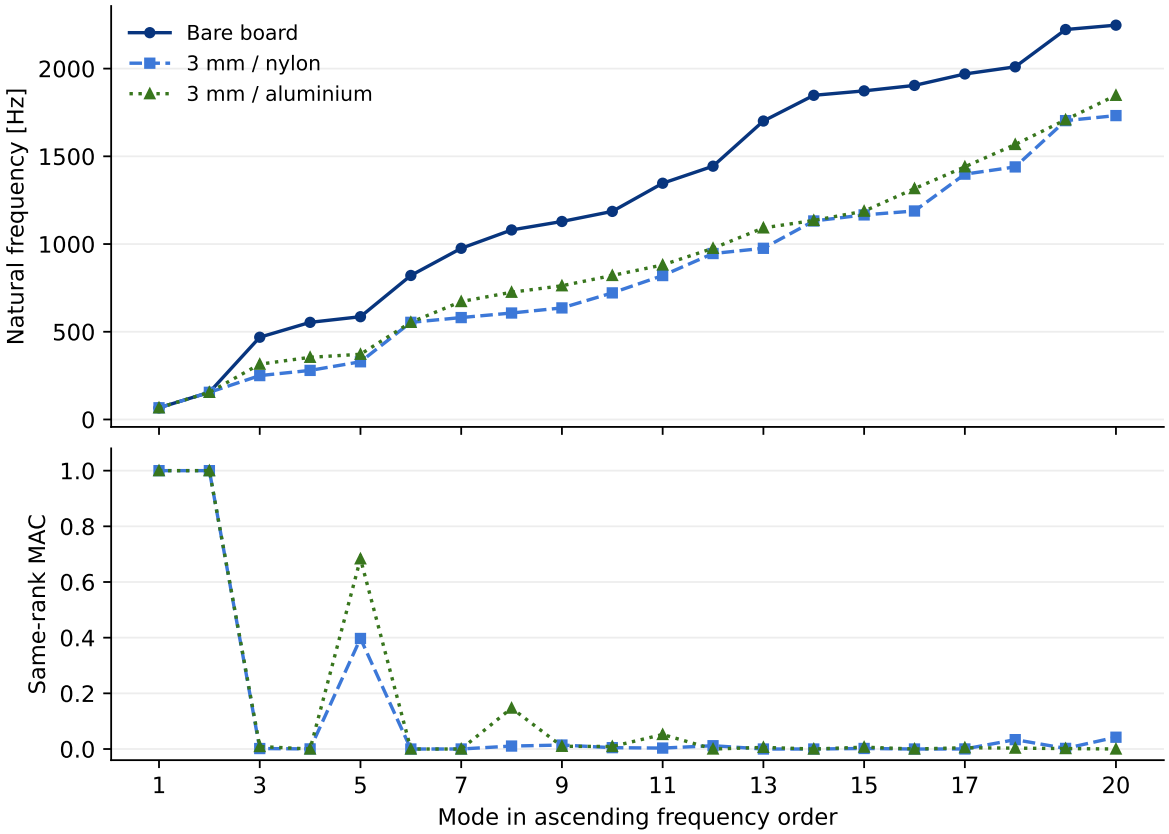


Figure 20: The upper panel orders each assembly's 20 native frequencies. The lower panel compares the bare and cased displacement vectors at equal rank. A value near one means that the original solid nodes moved in the same relative pattern.

Mounting	Mode 1 [Hz]	Mode 20 [Hz]	Mode 20 MAC to bare
Bare	66.611	2247.59	1.000
3 mm, nylon screws	66.614	1732.16	0.042
3 mm, aluminium screws	66.615	1847.75	< 0.001

The closest shape to bare mode 20 within either raised set was mode 19. Even those correlations were weak (0.129 for nylon and 0.137 for aluminium). The displayed aluminium mode 20 instead matches bare mode 14 at 1847.75 Hz (MAC 1.000). A matching shape for bare mode 20 may lie above the extracted range.

Reading the mode movies

Both default-contact movie pairs preserve the October mode ranks, a common colour range, and a 6 mm peak display amplitude. The raised panels identify the screw material, native frequency and shape correlation. These runs retain the original unpreloaded contacts. The tightened assembly has different equilibrium contact conditions and prestressed frequencies (page 8). Across the full catalogue, all 22 subjects now have a bare view and a comparison of both raised configurations. The flush assemblies still need an admissible installed state before their vibration fields can be shown.

The two cell attachments carry opposing lengthwise forces

The raised nylon case reduced the X force at both ends of all three cells under X excitation. Under Z excitation, however, both raised cases increased the force along the cell axis (Y) at all six attachments. The largest change occurred on the right-hand cell. The two ends carried 2.07 and 1.54 N RMS bare, 4.14 and 3.91 N with nylon screws, and 3.55 and 3.33 N with aluminium screws.

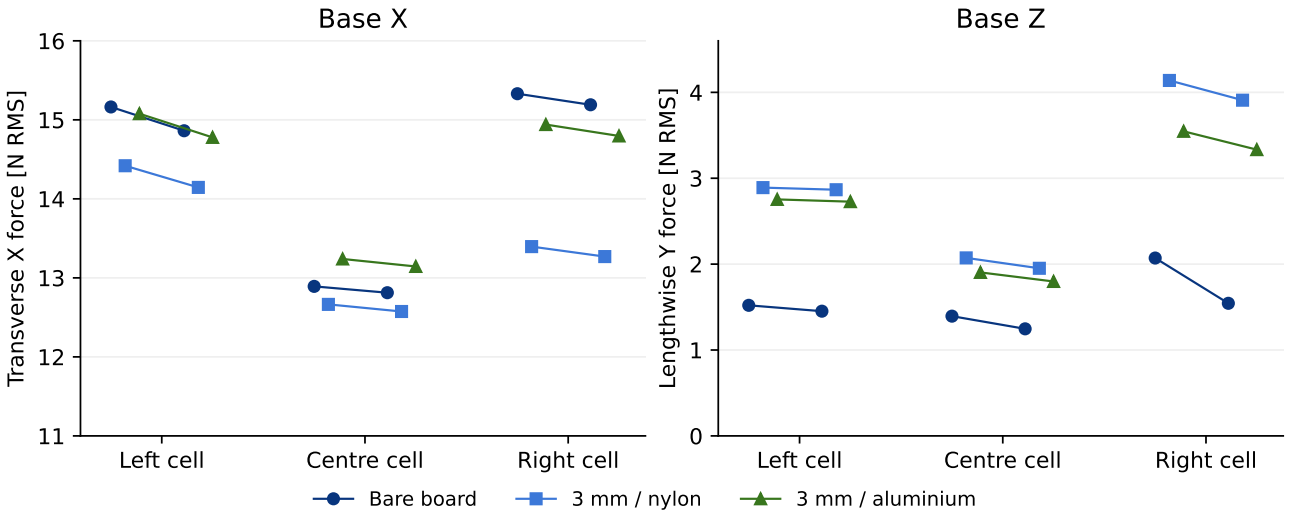


Figure 21: Each line joins the low-Y and high-Y ends of one cell. Left, the X force for base X. Right, the lengthwise Y force for base Z. Each marker gives the one-sigma force integrated over a complete cell-to-nickel attachment.

Why the whole-cell force can stay small

For the right-hand cell under base Z, the two lengthwise force histories had correlations of −0.88 bare, −0.97 with nylon and −0.97 with aluminium in the spectrum calculation. When the signed forces were added before taking RMS, the corresponding net loads were only 1.03, 0.99 and 0.90 N. The opposing forces matter to the individual attachments even when they nearly cancel across the cell.

Right cell, base X	Bare	Raised nylon	Raised aluminium
Low-Y end, X force [N RMS]	15.33	13.40	14.94
High-Y end, X force [N RMS]	15.19	13.27	14.80
Combined X force [N RMS]	30.52	26.66	29.73

Force recovery and checks

Both sides of each bond were identified from the native contact and target connectivity. For each tightened mode, the assembled solid-element stiffness and inertia forces cancel at free cell nodes. Summing the residuals on a bonded face gives its signed modal force. The adjacent nickel strip reproduced the opposite resultant to within 0.00032% in vector norm for every mode. Independent frequency integration then combined the six bond resultants over the original base spectrum with 2% modal damping.

For 90 sampled cell and nickel elements, the independently combined stiffness-force components agreed with all nine native PSD cases within 0.0056% in relative norm. Under base X the summed X loads agreed with the whole-cell inertia calculation (page 20) within 0.42%. Across Y and Z the corresponding differences reached 4.73% and 2.16%. Removing modes above 3.5 kHz changed any component by at most 0.362 N. The original October mode forces and native coordinates reproduced all 480 bare-board movie frames for all three drives within 0.00043 N per component.

The bonds cover the areas joined in the model. Actual weld sizes, local separation and the response after cell-to-case impact remain unmeasured.